

2. Variational Perspective: from VAEs to DDPMs

3w2 Thu 3/2 ~ 3w2 Thu 3/2

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2.1. Variational AutoEncoder 3w2 Thu 3/2

$$z \sim p_{prior} := N(0, I) \rightarrow x \sim p_{\phi}(x|z).$$

$\Rightarrow$ Conditional Generation $\frac{p(x|z)}{p(z)}$.

$$p_{\phi}(x) = \int_{\mathbb{R}^d} p_{\phi}(x|z) p_0(z) dz. \quad : \text{Marginal Likelihood.}$$

$\Rightarrow$ Intractable.

$\Rightarrow$ By Bayes Rule.

$$p_{\phi}(z|x) = \frac{p_{\phi}(x|z)p_0(z)}{p_{\phi}(x)} \rightarrow \text{Intractable.}$$

$\Rightarrow q_{\phi}(z|x) \approx p_{\phi}(z|x) : \text{Encoder.}$

Evidence Lower Bound

For any data point x

$$\log p_{\phi}(x) \geq \mathcal{L}_{ELBO}(\phi; x)$$

$$= \underbrace{\mathbb{E}_{z \sim q_{\phi}(x)} [\log p_{\phi}(x|z)]}_{\text{Reconstruction Term}} - \underbrace{\mathcal{D}_{KL}(q_{\phi}(x) \| p_0)}_{\text{Latent Regularization}}.$$

$$\text{pf. } \log p_{\phi}(x) = \log \int p_{\phi}(x|z) dz = \log \int q_{\phi}(z|x) \frac{p_{\phi}(x|z)}{q_{\phi}(z|x)} dz$$

$$= \log \mathbb{E}_{z \sim q_{\phi}(z|x)} \frac{p_{\phi}(x|z)}{q_{\phi}(z|x)} \geq \mathbb{E}_{z \sim q_{\phi}(z|x)} \log \frac{p_{\phi}(x|z)}{q_{\phi}(z|x)} \quad (\text{Jensen's Inequality})$$

$\Rightarrow$ Random Intermediate pdf for approximation.

$$= \mathbb{E}_{z \sim q_{\phi}(z|x)} \cdot \log p_{\phi}(x|z) + \underbrace{\mathbb{E}_{z \sim q_{\phi}(z|x)} \log \frac{p_0(z)}{q_{\phi}(z|x)}}_{= -\mathcal{D}_{KL}(q_{\phi}(z|x) \| p_0(z))}$$

$\Rightarrow$ Sampling from conditional dist. $q_{\phi}(z|x)$.

Computing only the conditional prob. $p_{\phi}(x|z)$.

Gaussian VAE

q_{ϕ} and p_{ϕ} must be computable.

$\Rightarrow$ Select Gaussian.

$$q_{\phi}(z|x) := N(z; \mu_{\phi}(x), \text{diag}(\sigma_{\phi}^2(x))). \quad \mu_{\phi}: \mathbb{R}^d \rightarrow \mathbb{R}^d \quad \sigma_{\phi}: \mathbb{R}^d \rightarrow \mathbb{R}_+^d$$

$$p_{\phi}(x|z) := N(x; \mu_{\phi}(z), \sigma^2 I). \quad \mu_{\phi}: \mathbb{R}^d \rightarrow \mathbb{R}^d \quad \sigma > 0$$

$\Rightarrow p_{\phi}$ is ~~intractable~~ $\frac{p_{\phi}(x|z)}{q_{\phi}(z|x)}$?

$\Rightarrow$ ELBO for Gaussian VAE.

$$\text{Reconstruction Term: } \mathbb{E}_{z \sim q_{\phi}(z|x)} \log p_{\phi}(x|z) = \mathbb{E}_{z \sim q_{\phi}(z|x)} \left[-\frac{1}{2\sigma^2} \|x - \mu_{\phi}(z)\|^2 \right]$$

$$\Rightarrow \min_{\phi} \mathbb{E}_{z \sim q_{\phi}(z|x)} \left[-\frac{1}{2\sigma^2} \|x - \mu_{\phi}(z)\|^2 \right] + \mathcal{D}_{KL}(q_{\phi}(z|x) \| p_0(z)).$$

$\Rightarrow$ Single peak? Averaging Effect.

2.2. DDPM 3w2 Thu 3/2

Forward Process (Fixed Encoder) $p(x_i|x_{i-1})$

Reverse Process (Learnable Decoder) $p_{\phi}(x_{i-1}|x_i)$.

$$p(x_i|x_{i-1}) := N(x_i; \sqrt{1-\beta_i} x_{i-1}, \beta_i^2 I) \quad \beta_i = \beta_{i-1}^L : \text{increasing noise sampler.}$$

$$\alpha_i = \sqrt{1-\beta_i^2}$$

$$\Rightarrow x_i = \alpha_i x_{i-1} + \beta_i \varepsilon_i \quad \varepsilon_i \sim N(0, I).$$

$$\mathbb{E} x_i = \alpha_i \mathbb{E} x_{i-1} + \beta_i \mathbb{E} \varepsilon_i = \alpha_i \mathbb{E} x_{i-1} = \left(\prod_{k=1}^i \alpha_k \right) x_0.$$

$$\text{Var } x_i = (1-\beta_i^2) \text{Var } x_{i-1} + \beta_i^2 = \prod_{k=1}^i (1-\beta_k^2) \text{Var } x_0 + \sum_{j=1}^i \beta_j^2 \prod_{k=j+1}^i (1-\beta_k^2)$$

$$= 1 - \prod_{k=1}^i \alpha_k^2$$

$$\therefore x_i = \bar{\alpha}_i x_0 + (1-\bar{\alpha}_i^2)^{\frac{1}{2}} \varepsilon \quad (\bar{\alpha}_i = \prod_{k=1}^i \alpha_k).$$

Since $\{\beta_i\}_{i=1}^L$ is an increasing noise scheduler,

$$\frac{p_L(x_L|x_0) \rightarrow N(0, I)}{\Rightarrow p_{prior} = N(0, I)} \quad n$$

We learn the REVERSE PROCESS of this predetermined encoder.

$$\text{We know that } p(x_i|x_0) = N(x_i; \bar{\alpha}_i x_0 + (1-\bar{\alpha}_i^2)^{\frac{1}{2}} \varepsilon).$$

$$\text{We want to know } p(x_0|x_L), \text{ or } p(x_{i-1}|x_i).$$

$$\text{Therefore, our goal is now } \mathbb{E}_{x_i \sim p_i(x_i)} [\mathcal{D}_{KL}(p(x_{i-1}|x_i) \| p_{\phi}(x_{i-1}|x_i))]$$

We can sample from $x_i \sim p_i(x_i)$ with conditional generation from data samples

$$\text{However, } p(x_{i-1}|x_i) = p(x_i|x_{i-1}) \frac{p_{\phi}(x_{i-1})}{p_0(x_i)} \Rightarrow \text{Intractable.}$$

$$p(x_i) = \int_{\mathbb{R}^d} p(x_i|x_0) p_{prior}(x_0) dx_0.$$

$\Rightarrow$ Data sample로 $\frac{p_{\phi}(x_{i-1})}{p_0(x_i)}$ 값을 구할 수 있는가? 아니면 $\frac{p_{\phi}(x_{i-1})}{p_0(x_i)}$ 값을 구하는 것?

1. Blurry Problem (Matching EM w/ Single Gaussian)

$\Rightarrow$ Multiple peaks.

$\Rightarrow$ Model parameterization.

$\Rightarrow$ $\frac{p_{\phi}(x_{i-1})}{p_0(x_i)}$ 값을 구할 수 있는가? (ELBO는 x_i (noise)에 x_i (noise)를 더함?)

$\Rightarrow$ Generative model

2. We can't use more batch $\rightarrow$ data only.

Overcoming Intractability with conditioning

$$p(x_{i-1}|x_i, x_0) = p(x_i|x_{i-1}) \frac{p(x_{i-1})}{p_0(x_i)}$$

$\Rightarrow$ Conditioning of Data Sample로 $\frac{p(x_{i-1})}{p_0(x_i)}$ 값을 구할 수 있는가?

Equivalence Between marginal and conditional KL minimization.

$$\mathbb{E}_{x_i \sim p_i(x_i)} [\mathcal{D}_{KL}(p(x_{i-1}|x_i) \| p_{\phi}(x_{i-1}|x_i))]$$

$$= \mathbb{E}_{x_0 \sim p_{prior}(x_0)} \mathbb{E}_{x_i \sim p_i(x_i|x_0)} [\mathcal{D}_{KL}(p(x_{i-1}|x_i, x_0) \| p_{\phi}(x_{i-1}|x_i, x_0))].$$

PROOF?

Reverse Conditional Transition Kernel

$$p(x_{i-1}|x_i, x_0) = N(x_{i-1}; \frac{\bar{\alpha}_i \beta_i^2}{1-\bar{\alpha}_i^2} x_0 + \frac{(1-\bar{\alpha}_i^2)}{1-\bar{\alpha}_i^2} \alpha_i x_i, \frac{1-\bar{\alpha}_i^2}{1-\bar{\alpha}_i^2} \beta_i^2)$$

PROOF?

?

Modeling of Reverse Transition kernel $p_{\phi}(x_{i-1}|x_i)$

Now that we know the closed form of $p(x_{i-1}|x_i, x_0)$, Equivalence between conditioned & Marginal Loss.

$$\text{Let. } p_{\phi}(x_{i-1}|x_i) := N(x_{i-1}; \mu_{\phi}(x_i, i), \sigma^2 I).$$

$$\mathcal{L}_{prior}(\phi) = \mathbb{E}_{x_0 \sim p_{prior}(x_0)} \sum_{i=1}^L \mathbb{E}_{x_i \sim p_i(x_i|x_0)} [\mathcal{D}_{KL}(p(x_{i-1}|x_i, x_0) \| p_{\phi}(x_{i-1}|x_i, x_0))]$$

$$= \mathbb{E}_{x_0 \sim p_{prior}(x_0)} \sum_{i=1}^L \mathbb{E}_{x_i \sim p_i(x_i|x_0)} \underbrace{\mathbb{E}_{x_{i-1} \sim p(x_{i-1}|x_i, x_0)} \left[\frac{1}{2\sigma^2} \|\mu_{\phi}(x_i, i) - \mu(x_i, x_0, i)\|^2 \right]}_{\Rightarrow \text{prior}}$$

$$= \sum_{i=1}^L \mathbb{E}_{p(x_0)} \frac{1}{2\sigma^2} \|\mu_{\phi}(x_i, i) - \mu(x_i, x_0, i)\|^2$$

2 choices of parameterization.

$$\varepsilon\text{-prediction. Let } x_i = \bar{\alpha}_i x_0 + \sqrt{1-\bar{\alpha}_i^2} \varepsilon \quad (\varepsilon \sim N(0, I)).$$

$$\mu(x_i, x_0, i) = \frac{1}{\alpha_i} \left(x_i - \frac{1-\alpha_i^2}{\sqrt{1-\bar{\alpha}_i^2}} \varepsilon \right) \quad (\text{Reverse kernel})$$

$\Rightarrow$ parameterize μ_{ϕ} as ε_{ϕ} .

$$\|\mu_{\phi} - \mu\|^2 \propto \|\varepsilon_{\phi} - \varepsilon\|^2.$$

$\Rightarrow$ Loss Simplification.

$$\mathcal{L}_{simple} = \mathbb{E}_{i, x_0 \sim p_{prior}(x_0), \varepsilon \sim N(0, I)} [\|\varepsilon_{\phi}(x_i, i) - \varepsilon\|^2].$$

$$x\text{-prediction. } \mu_{\phi}(x_i, x_0, i) = \frac{\bar{\alpha}_i \beta_i^2}{1-\bar{\alpha}_i^2} x_0 + \frac{(1-\bar{\alpha}_i^2)}{1-\bar{\alpha}_i^2} \alpha_i x_i$$

$\Rightarrow$ Predict the clean sample.

DDPM's ELBO

$$-\log p_{\phi}(x_0) \leq -\mathcal{L}_{ELBO}(x_0; \phi).$$

$$:= \mathcal{L}_{prior}(x_0) + \mathcal{L}_{den}(x_0; \phi) + \mathcal{L}_{prior}(x_0; \phi).$$